

## Experience

### Beyond Cracking the Coding Interview

Author

2025

- Co-wrote the official sequel to Cracking the Coding Interview with Gayle L. McDowell et al. #1 Best Seller in Data Structures & Algorithms on Amazon ([amazon.com/dp/195570600X](https://amazon.com/dp/195570600X)).

### Google

Senior Software Engineer

Nov 2022 – Aug 2024

- Part of a team that migrated the bandwidth ordering system from a quarterly manual process to an on-demand service with daily resolution, giving internal teams more flexibility and improving WAN utilization. (C++, Go, Python, SQL)
- Built and owned the bandwidth allocation and validation components. Designed time-series-based algorithms for distributing the projected available bandwidth among products, balancing prioritization, fairness, and continuity constraints when demand exceeds capacity.
- Point of contact in a cross-functional effort to rapidly provision network capacity for Gemini. Quickly prototyped and then standardized a new mechanism for specifying and prioritizing business-critical traffic while minimizing impact to other traffic.

Software Engineer

Feb 2021 – Nov 2022

- Designed a rollout scheduler for config changes to Google's global WAN (Wide Area Network), parallelizing deployments across continents to halve deploy time while reducing outage risk.

## Education

PhD + Master's in Computer Science University of California Irvine, GPA 3.83/4

Sep 2015 – Dec 2019

- Co-authored 9 peer-reviewed papers on algorithm design, including as main author in tier A conferences like ICALP and ISAAC. The papers describe new algorithmic improvements in graph theory (routing), computational geometry (clustering, matching, TSP), and computational biology (network alignment) ([scholar.google.bg/citations?user=LLuligEAAAAJ](https://scholar.google.bg/citations?user=LLuligEAAAAJ)).
- Led a research project from inception to publication: came up with an original problem, engaged 3 colleagues to work on it, and collaborated with them to solve it and write a paper. We invented an algorithm for the knight's tour problem (cited by Knuth).
- Led a 120-student study on the effect of immediate automated feedback on learning outcomes.

B.E. in Computer Science Polytechnic University of Catalonia, GPA 3.8/4 (99th percentile)

Sep 2011 – Jul 2015

- Created SANA, a C++ biological network alignment tool that uses simulated annealing. 100+ citations; actively maintained for 10+ years by 50+ collaborators, with 2,000+ commits ([github.com/nmamano/SANA](https://github.com/nmamano/SANA)).

## Projects

Isomux [isomux.com](https://isomux.com)

2026

- Built an agent orchestration system where conversations are multi-agent, multi-user, multi-provider, and multi-device. (React, Bun, WebSocket, Claude Agent SDK, Codex App Server)
- Isometric office UI with animated agent workers as an intuitive visual metaphor.
- Bun process manages agent lifecycles, inter-agent messaging, and real-time sync across clients via WebSocket.

DSA Toolkit [dsatoolkit.com](https://dsatoolkit.com)

2025

- A companion DS&A toolkit for Beyond Cracking the Coding Interview. Built an automated pipeline to parse, test, and generate articles for all 1,440 code solutions across 5 languages. (Next.js, TypeScript, Python)
- Built LLM orchestration pipelines for large-scale codebase changes, e.g., translating 288 book solutions from Python to Go with few-shot prompting and test-driven retry loops, using local models and APIs.
- Co-designed an AI interviewer for the book problems, with tens of thousands of interviews completed ([bctci.co/ai](https://bctci.co/ai)).

Wall Game [wallgame.io](https://wallgame.io)

2025

- A full-stack online board game supporting AIs as first-class citizens via an AI integration protocol/client. (React, TypeScript, Tailwind, shadcn, PostgreSQL, MongoDB, WebSocket)
- Trained multiple AlphaZero-style models for different board sizes and rule sets via self-play on consumer GPUs. (C++, PyTorch, TensorRT, ONNX, WebAssembly)

Technical Blog [nilmamano.com/blog](https://nilmamano.com/blog)

2025

- 30+ articles on algorithms, data structures, AI, CS research, and software engineering.